For the accidental background calculation you have to calculate the expected background rate in the detector to compare it with you neutrino rate per day :

accidental background rate = prompt event rate * delayed event rate * coincidence time (70 microseconds)

let's make the approximations that the prompt and the delayed are just gammas, so we have for the prompt events:
prompt event rate (gamma) = gamma rate/day * efficiency of gamma detection as a prompt [2-10 MeV]
where
gamma rate/day = gamma/cm2/s (from literature) * surface* number of seconds in one day
and
efficiency of gamma detection as a prompt = number of detected gamma events in histogram between 2 and 10 MeV / number of simulated gammas

and for the delayed events :
the delayed event rate(gamma) = gamma rate/day * efficiency of gamma detection as a delayed [5-10 MeV]
where
neutron rate/day = gammas/cm2/s (from literature) * surface* number of seconds in one day
and
efficiency of gamma detection as a delayed = number of detected gamma events in histogram between 5 and 10 MeV / number of simulated gammas

then you can compare this with your neutrino signal per day and make a signal/noise ratio.

This ratio should be equal or bigger than 1.